香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

ECO6102 / 2026–27 Term 1 (Fall)

Lecture 4b: Instrumental Variables

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Lecture 4 · Mondays 12 and 19 October 2026

Midterm arrangements and Problem Set 2

Prepare the permitted sheet and finish the IV practice

- **Midterm** Monday 26 October 2026, 09:00–11:00; 西丽小礼堂 (tentative). Covers L1–L4; 30% of the course grade.
- **Format** closed book and in person; one double-sided A4 cheat sheet is allowed. No calculators or electronic dictionaries.
- **Problem Set 2** Due Friday 23 October 2026. Solutions will be released immediately after the deadline.

Exam venue map

Contents

Introductory Econometrics 15-3–15-6

1. Two Stage Least Squares
2. IV Solutions to Errors-in-Variables Problems
3. Testing for Endogeneity and Overidentifying Restrictions
4. 2SLS with Heteroskedasticity

Technical extensions are linked in the appendix; Midterm Review is a separate deck.

Continue with two stages and diagnostics for their estimates.

Recap: IV

Moments connect the assumptions to the estimator

$$y = \beta_0 + \beta_1 x + u, \qquad E(u) = 0.$$

$$\text{Cov}(z, u) = 0, \qquad \text{Cov}(z, x) \neq 0.$$

$$\text{Cov}(z, y) = \beta_1 \text{Cov}(z, x) \implies \hat{\beta}_{1,IV} = \frac{\sum_i (z_i - \bar{z})(y_i - \bar{y})}{\sum_i (z_i - \bar{z})(x_i - \bar{x})}.$$

Exogenous variables Potentially endogenous regressor Structural error

Recall IV with included exogenous variables

Exogeneity is maintained; relevance must remain after the included variables

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + u_1.$$

$$E(u_1) = 0, \quad \text{Cov}(exper, u_1) = \text{Cov}(fatheduc, u_1) = 0.$$

$$educ = \pi_0 + \pi_1 exper + \pi_2 fatheduc + v_2, \qquad \pi_2 \neq 0.$$

A second instrument needs its own argument

Relevance and exogeneity are separate requirements

Same target

$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + u_1$.

Mother's education also predicts schooling. What additional assumption is needed for it to be a valid IV?

$$\text{Cov}(motheduc, u_1) = 0.$$

If both parents are credible instruments, how should we use their information together?

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Intro 15-3a

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We have several credible instruments. How can we combine their information?

Start with the model and the excluded variables

One endogenous regressor; two candidates excluded from the outcome equation

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1. \tag{15.22}$$

Already included z_1 has its own effect on y_1. We maintain

$$E(u_1) = 0, \qquad \text{Cov}(z_1, u_1) = 0.$$

Now add two candidates z_2, z_3 do not enter this equation, and

$$\text{Cov}(z_2, u_1) = 0, \qquad \text{Cov}(z_3, u_1) = 0.$$

These are the exclusion restrictions.

Relevance Together, the excluded variables must predict y_2 beyond the included z_1.

Why combine the instruments?

Using each instrument separately leaves information unused

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1.$$

Using either valid instrument separately generally leaves information unused; the two IV estimators need not be efficient. A fixed combination remains uncorrelated with u_1:

$$\mathrm{Cov}(a_0 + a_1 z_1 + a_2 z_2 + a_3 z_3, u_1) = \sum_{j=1}^{3} a_j \,\mathrm{Cov}(z_j, u_1) = 0.$$

Exogeneity survives. Which linear combination is most strongly correlated with y_2?

Why combinations remain exogenous

Choose the most strongly correlated combination

The reduced form supplies the best linear prediction

Structural model

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1$; all z_j are maintained exogenous.

Choose the linear combination most highly correlated with y_2.

$$y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + \pi_3 z_3 + v_2. \tag{15.33}$$

The population linear projection defines its remainder by

$$E(v_2) = 0, \qquad \mathrm{Cov}(z_j, v_2) = 0 \quad (j = 1, 2, 3).$$

The reduced form gives the instrument

Combine the exogenous variables using their population coefficients

Reduced form

$y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + \pi_3 z_3 + v_2$.

$$y_2^* = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + \pi_3 z_3. \tag{15.34}$$

Use the best linear prediction of the endogenous regressor as its instrument.

It is the best IV among linear combinations under the assumptions in Appendix15A, including homoskedasticity.

Why this prediction maximizes correlation

The excluded variables must add prediction

Relevance is prediction beyond the included exogenous variable

Structural equation

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1$.

First stage

$y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + \pi_3 z_3 + v_2$.

$$\pi_2 \neq 0 \quad \text{or} \quad \pi_3 \neq 0. \tag{15.35}$$

If both are zero, $y_2^* = \pi_0 + \pi_1 z_1$. There is no separate predicted variation to identify the effect of y_2 while holding z_1 fixed.

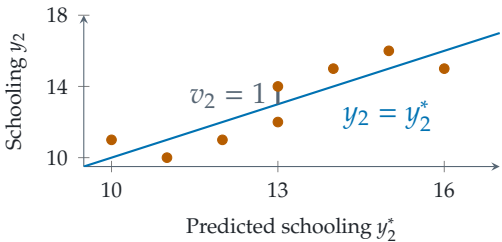
The population moment explanation

Prediction and the schooling remainder

15-3a: the first-stage remainder is a vertical gap

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1$, with valid IVs.

$$y_2 = y_2^* + v_2. \tag{15.33}$$



One schooling value

$14 = 13 + 1$.

$\mathrm{Cov}(y_2^*, u_1) = 0$ follows from IV exogeneity.

The remainder v_2 may share u_1. This is where endogeneity can remain.

Population to sample

The sample estimates the same projection rule

The population coefficients π_j are unknown. Estimate the reduced form by OLS:

$$\widehat{y_2} = \hat{\pi}_0 + \hat{\pi}_1 z_1 + \hat{\pi}_2 z_2 + \hat{\pi}_3 z_3. \tag{15.36}$$

Check excluded-variable relevance: test $H_0 : \pi_2 = \pi_3 = 0$.

This checks prediction, not exogeneity. Instrument strength follows in15-3c.

From population projection to sample OLS

Use the fitted value as the instrument

Three equations jointly determine the IV coefficients

Structural residual

$\hat{u}_{i1} = y_{i1} - \hat{\beta}_0 - \hat{\beta}_1 y_{i2} - \hat{\beta}_2 z_{i1}$.

As in 15-2, (15.25), use $\widehat{y_{i2}}$ in place of the single excluded IV:

$$\sum_i \hat{u}_{i1} = 0,$$

$$\sum_i z_{i1} \hat{u}_{i1} = 0,$$

$$\sum_i \widehat{y_{i2}} \hat{u}_{i1} = 0. \tag{15.37}$$

Population and sample moments

Compute the IV coefficients in two stages

OLS on the fitted regressor gives the same IV coefficients

Using $\widehat{y_2}$ as the IV for y_2 gives the same coefficients as the second regression below.
Stage 1 Regress y_2 on z_1, z_2, z_3 and an intercept:

$$\widehat{y_2} = \hat{\pi}_0 + \hat{\pi}_1 z_1 + \hat{\pi}_2 z_2 + \hat{\pi}_3 z_3. \tag{15.36}$$

Stage 2 Run the OLS regression

$$y_1 \text{ on } \widehat{y_2}, z_1 \text{ and an intercept.} \tag{15.38}$$

This is two-stage least squares (2SLS). Retain the same sample and included controls in both stages.

Why the coefficients coincide

Why the predicted component works

Substitution changes the error but keeps the structural coefficient

Structural model

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1$.

Split $y_2 = y_2^* + v_2$: prediction plus remainder. Substitute into the structural equation:

$$y_1 = \beta_0 + \beta_1 y_2^* + \beta_2 z_1 + \underbrace{u_1 + \beta_1 v_2}_{\text{composite error}} . \tag{15.39}$$

The composite error has mean zero and is uncorrelated with y_2^* and z_1. This is why the population regression retains β_1.

Why both covariances are zero

Use IV software for standard errors

The two regressions compute coefficients but not ordinary OLS uncertainty

Population explanation

$y_1 = \beta_0 + \beta_1 y_2^* + \beta_2 z_1 + (u_1 + \beta_1 v_2)$.

- **Different error** The transformed error contains v_2; IV uncertainty uses the structural error u_1.
- **Different standard errors** Ordinary second-stage OLS standard errors and tests are not valid for 2SLS.
- **Use IV software** Supply the actual outcome, actual endogenous regressor, included controls and complete IV list to a 2SLS command.

Residuals and standard errors | Stata | R | Python

One instrument is a special case of 2SLS

First connect to the father-only model; then combine both parents

Recap: Example 15.1

$\log(wage) = \beta_0 + \beta_1 educ + u$; one excluded IV: *fatheduc*.

Calculation on the same 428 women	Schooling slope
Direct IV ratio	.05917
Two-stage OLS using fitted schooling	.05917

With one instrument, IV and 2SLS coincide. The optional pictures show the same calculation. Next add experience controls and both parents, as in Example 15.5.

Why the coefficients are equal | The two-stage pictures

Combine both parents and retain experience controls

Example 15.5: one prediction for the same structural schooling effect

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u_1. \tag{15.40}$$

- **Potential endogeneity** Ability remains in u_1.
- **Included controls** Experience and its square enter both stages and are maintained exogenous.
- **Excluded instruments** Both parents' education predict schooling. We assume no remaining wage path through ability, connections or family resources: $\text{Cov}(motheduc, u_1) = \text{Cov}(fatheduc, u_1) = 0$.

Use the same 428 women for the OLS–2SLS comparison.

First stage combines both parents

Example 15.5: test what the excluded parents add beyond experience

$$educ = \pi_0 + \pi_1 exper + \pi_2 exper^2 + \pi_3 motheduc + \pi_4 fatheduc + v_2. \tag{15.41}$$

$$H_0 : \pi_3 = \pi_4 = 0, \qquad F(2, 423) = 55.40 \quad (n = 428).$$

The parents jointly add schooling prediction beyond the included controls. This is relevance evidence.

Fit (15.41) by OLS to obtain $\widehat{educ}_i$, the schooling prediction used next.

Calculate one woman's predicted schooling

Two stages give a 6.1 percent estimate

Example 15.5: read the fitted-schooling slope

Structural target

$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u_1.$

Stage 2 Regress log wages on fitted schooling and the included experience terms:

$$\log(wage_i) \text{ on } \widehat{educ}_i,\ exper_i,\ exper_i^2, \quad \text{and an intercept.}$$

$$\hat{\beta}_{1,2SLS} = .06140.$$

Holding experience fixed, an additional schooling year is associated with about 6.1% higher wages under the maintained IV assumptions.

Compare OLS and IV on the same model and sample

Example 15.5: the estimate difference needs an economic explanation

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u_1.$$

Same 428 women and model	OLS	2SLS
Schooling estimate	.1075	.0614
Conventional SE	.0141	.0314

Smaller IV is consistent with upward ability bias if the IVs are valid.

Why is the 2SLS estimate less precise? Next, examine the variation left to estimate its slope.

Omitted ability: diagram and bias formula

Brief note on consistency and assumptions

Consistency and conventional inference require different assumptions

Structural model

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1$.

- **Consistency** Maintain $E(u_1) = 0$, $\text{Cov}(z_j, u_1) = 0$ for every IV and excluded partial relevance $\pi_2 \neq 0$ or $\pi_3 \neq 0$.
- **Asymptotically valid standard errors** Also require the homoskedasticity condition

$$E(u_1^2 \mid \mathbf{z}) = \sigma^2.$$

This is the conventional variance formula; robust standard errors allow heteroskedasticity.

Full 2SLS assumptions

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We have a two-stage estimate. Why can its standard error be much larger than OLS's?

Two changes affect precision

Separate the familiar control-overlap cost from instrument prediction

OLS benchmark Exogenous regressors; homoskedastic errors.

$$\text{Simple regression:} \quad \text{Avar}(\hat{\beta}_{1,OLS}) \approx \frac{\sigma^2}{SST_2},$$

$$\text{Multiple regression:} \quad \text{Avar}(\hat{\beta}_{1,OLS}) \approx \frac{\sigma^2}{SST_2(1-R_2^2)}.$$

R_2^2: regress y_2 on $1, z_1$. At fixed noise and spread, overlap with the other included regressor reduces precision.

Next, 2SLS uses predicted $\widehat{y_2}$: a second loss of variation.

GPA and partialling-out illustration

Use separate fitted variation in stage two

From OLS to fitted schooling: remove the included-variable prediction

Second stage
Regress y_1 on $\widehat{y_2}$, z_1 and an intercept.

$$\text{Avar}(\hat{\beta}_{1,2SLS}) \approx \frac{\sigma^2}{\widehat{SST}_2(1-\widehat{R}_2^2)}. \tag{15.43}$$

$\sigma^2 = \text{Var}(u_1)$ is the structural-error variance. $\widehat{SST}_2 = \sum_i(\widehat{y}_{i2} - \bar{\widehat{y}}_2)^2$: spread in fitted $\widehat{y_2}$.

$$\widehat{y}_{i2} = \hat{a}_0 + \hat{a}_1 z_{i1} + \hat{r}_i, \qquad \widehat{R}_2^2 \text{ is this regression's fit.}$$

A high $\widehat{R}_2^2$ leaves little fitted variation to distinguish β_1. This is not the first-stage R^2.

Why 2SLS can have a larger standard error

Compare the homoskedastic benchmark with exogenous regressors

$$\text{Avar}(\hat{\beta}_{1,OLS}) \approx \frac{\sigma^2}{SST_2(1-R_2^2)}, \qquad \text{Avar}(\hat{\beta}_{1,2SLS}) \approx \frac{\sigma^2}{\widehat{SST}_2(1-\widehat{R}_2^2)}.$$

Less total spread The first stage decomposes $y_2 = y_2^* + v_2$, with orthogonal pieces:

$$\text{Var}(y_2) = \text{Var}(y_2^*) + \text{Var}(v_2).$$

Less separate spread The included variables may explain much of $\widehat{y_2}$, leaving little for its own coefficient.

The sample sums-of-squares identity

The first stage separates explained and residual spread

The sums-of-squares identity makes the lost variation visible

OLS first stage with an intercept

$y_{i2} = \widehat{y}_{i2} + \hat{v}_{i2}$.

$$\underbrace{\sum_i (y_{i2} - \bar{y}_2)^2}_{SST} = \underbrace{\sum_i (\widehat{y}_{i2} - \bar{y}_2)^2}_{SSE\text{: explained}} + \underbrace{\sum_i \hat{v}_{i2}^2}_{SSR\text{: residual}} .$$

Thus $\widehat{SST}_2 = SSE \leq SST_2$: fitted schooling cannot contain more spread than actual schooling. Included regressors then absorb part of this fitted spread too. Under exogeneity and homoskedasticity, OLS uses the available variation more efficiently.

College proximity and precision

Example 15.4: little separate schooling variation

$$\log(wage) = \beta_0 + \beta_1 educ + \mathbf{c}'\gamma + u.$$

- **Sample** The same 3,010 men in each regression.
- **Excluded IV** *nearc4*: lived near a four-year college in 1966.
- **Included exogenous variables c** Experience and its square, black, South, current and 1966 SMSA, and 1966 regional indicators.

$$educ = \pi_0 + \pi_{near} nearc4 + \mathbf{c}'\pi_c + v.$$

Nearby colleges predict schooling after the same included variables: $\hat{\pi}_{near} = .320$, $SE = .088$, $t = 3.64$.

The 2SLS schooling estimate is larger but less precise

Example 15.4: compare the same model and sample

Same target

$\log(wage) = \beta_0 + \beta_1 educ + \mathbf{c}'\gamma + u$; excluded instrument: $nearc4$.

Table 15.1	OLS	2SLS
Schooling coefficient	.075	.132
Standard error	.003	.055

The much larger 2SLS standard error raises the next question: how much fitted schooling variation remains after **c**?

How much schooling variation is explained?

Example 15.4: compare the regressions and their fit

c: the same controls as in the wage regression.

$$educ_i = \hat{a}_0 + \mathbf{c}_i'\hat{\mathbf{a}} + \hat{e}_i, \qquad R_c^2 = .4748.$$

$$educ_i = \underbrace{\hat{\pi}_0 + \mathbf{c}_i'\hat{\boldsymbol{\pi}}_c + \hat{\pi}_n nearc4_i}_{\widehat{educ_i}} + \hat{v}_{i2}, \qquad R_{FS}^2 = .4771.$$

$$\widehat{educ_i} = \hat{b}_0 + \mathbf{c}_i'\hat{\mathbf{b}} + \hat{r}_i, \qquad \widehat{R}_2^2 = .9951.$$

The controls explain 99.51% of fitted schooling variation; only 0.49% remains for its separate effect.

Only separate fitted variation identifies the slope

Example 15.4: fitted spread times its remaining share

Second stage

$\log(wage)$ on $1, \mathbf{c}, \widehat{educ}$.

$$\text{Avar}(\hat{\beta}_{1,2SLS}) \approx \frac{\sigma^2}{\widehat{SST}_2(1 - \widehat{R}_2^2)}. \tag{15.43}$$

Separate fitted spread $10{,}287.62 \times (1 - .9951478) \simeq 49.92$.

Regress $\widehat{educ}$ on $1, \mathbf{c}$, excluding *nearc4*. Its small residual spread is the variance denominator.

Validity and precision answer different questions

A credible identifying direction must also contain enough variation

Validity Exogeneity and relevance determine whether the IV variation can identify the causal effect.

Precision Strength and the amount of separate fitted variation determine how precisely it does so. Collinearity can be especially costly after prediction has already reduced the spread.

Since IV can cost precision, we need to check its relevance and ask whether endogeneity requires it at all.

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Limited fitted variation reduces precision. Is first-stage significance enough for reliable inference?

What is a weak instrument?

The excluded variables predict little additional regressor variation

First stage

$y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + \pi_3 z_3 + v_2$.

Weak instruments have little predictive power for y_2 after the included exogenous variable z_1.

A weak first stage leaves little instrument-predicted variation for estimating the structural slope.

Recall why weak relevance matters

A small first-stage denominator magnifies a small violation

One-instrument recall

$y = \beta_0 + \beta_1 x + u$, with candidate instrument z.

$$\operatorname{plim} \hat{\beta}_{IV} = \beta_1 + \frac{\operatorname{Cov}(z, u)}{\operatorname{Cov}(z, x)}.$$

If $\operatorname{Cov}(z, x)$ is close to zero, even a small nonzero $\operatorname{Cov}(z, u)$ can cause a large discrepancy.

Test the excluded coefficients

Compare prediction with and without the excluded variables

Unrestricted first stage

$$educ = \pi_0 + \pi_1 exper + \pi_2 exper^2 + \pi_3 motheduc + \pi_4 fatheduc + v_2.$$

$$H_0 : \pi_3 = \pi_4 = 0.$$

Restricted first stage

$$educ = \pi_0 + \pi_1 exper + \pi_2 exper^2 + v_R.$$

The restricted model removes only the parents. Does adding them improve prediction beyond experience?

We use the same 428 observations as the outcome equation. An overall first-stage F would ask a different question.

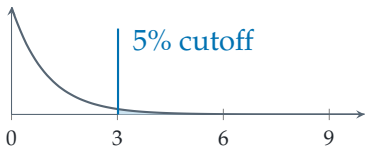
The first-stage F compares prediction errors

A large reduction in SSR points away from the null

Same test

$H_0 : \pi_3 = \pi_4 = 0$; restricted excludes parents, unrestricted includes them.

$$F = \frac{(SSR_R - SSR_{UR})/2}{SSR_{UR}/423}.$$



$F(2, 423)$ under H_0

Same-sample MROZ

$F = 55.40$, far to the right of the plotted range.

Reject: the parents jointly add predictive power after experience.

One excluded instrument: use a t test

The same single-coefficient test gives F equal to t squared

One excluded IV

$y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + v_2$; included: z_1, excluded: z_2.

$$H_0 : \pi_2 = 0, \qquad F = t^2_{\pi_2}.$$

A large $|t|$ (one excluded IV), or a large F (several), is evidence against zero excluded-variable relevance.

Rejecting zero relevance is not enough to rule out weak instruments; the stronger screening guideline follows next.

Common threshold

Strength requires more than the usual significance cutoff

Same first stage
Test the excluded instruments after all included exogenous variables.

$$F > 10, \qquad |t| > \sqrt{10} \simeq 3.2 \text{ for one excluded IV.}$$

Weak IV - testing relevance

State what the first-stage evidence establishes

- State the first-stage equation and identify the excluded coefficients.
- Read the t or joint F and the strength guideline in the context of the design.
- Keep relevance evidence separate from the instrument-exogeneity argument.

Several endogenous variables: optional extension | Optional coefficient testing

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IV can avoid omitted-variable bias. Can it also avoid noise in a measured regressor?

A new source of endogeneity: measurement error

15-4: IV can address omitted causes and noise in a regressor

Identification threat

$y = \beta_0 + \beta_1 x + u$, with $\text{Cov}(x, u) \neq 0$.

- **Omitted variables** A regressor shares an unobserved cause with the outcome. We used IV to isolate variation avoiding that cause.
- **Measurement error** The observed value differs from the true value. Does it affect the outcome or a regressor?
- Classical noise in y adds uncertainty. Classical noise in x can create endogeneity; a valid IV can help.

Noise in the outcome need not bias the slope

From omitted variables to measurement error: start with outcome noise

True model

$y^* = \beta_0 + \beta_1 x + u$, with $\text{Cov}(x, u) = 0$.

$$y^{obs} = y^* + e_y = \beta_0 + \beta_1 x + (u + e_y).$$

If $\text{Cov}(x, e_y) = 0$, then

$$\text{plim}\, \hat{\beta}_{1,OLS} - \beta_1 = \frac{\text{Cov}(x, u + e_y)}{\text{Var}(x)} = 0.$$

Independent outcome noise increases error variance and reduces precision. Systematic measurement error related to x can also bias the slope.

Start from the true regressor and its observed measure

Equation 15.45 defines what is measured with error

$$y = \beta_0 + \beta_1 x_1^* + \beta_2 x_2 + u. \tag{15.45}$$

$$x_1 = x_1^* + e_1.$$

The classical measurement-error assumptions give

$$\text{Cov}(e_1, x_1^*) = \text{Cov}(e_1, x_2) = \text{Cov}(e_1, u) = 0.$$

The true regressors are uncorrelated with u; the measured regressor contains additional noise.

Measurement noise enters the regression error

The measured regressor and transformed error share the same noise

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + (u - \beta_1 e_1). \tag{15.46}$$

Under the classical assumptions,

$$\operatorname{Cov}(x_1, u - \beta_1 e_1) = -\beta_1 \operatorname{Var}(e_1).$$

Positive measurement-error variance creates endogeneity when $\beta_1 \neq 0$. In the classical case the OLS slope is biased toward zero.

Noise in a regressor attenuates its slope

Classical measurement error adds variation without adding signal

Simple model: omit the control

$y = \beta_0 + \beta_1 x_1^* + u$, observed $x_1 = x_1^* + e_1$.

$$\operatorname{plim} \hat{\beta}_{1,OLS} = \beta_1 \underbrace{\frac{\operatorname{Var}(x_1^*)}{\operatorname{Var}(x_1^*) + \operatorname{Var}(e_1)}}_{\text{signal share}}.$$

Under classical noise, the signal share lies between zero and one: the slope moves toward zero (attenuation). More data does not fix this bias; a valid IV can.

Deriving attenuation

Two IV remedies avoid the same measurement noise

Use the shared signal while avoiding the transformed outcome error

Observed-regressor equation

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + (u - \beta_1 e_1)$, where $x_1 = x_1^* + e_1$.

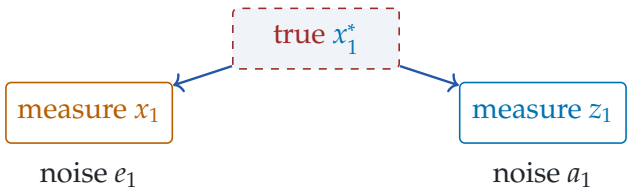
1. **Another measure** Report the same true variable through a second measurement process. Separate sources of error can make uncorrelated measurement noise plausible.
2. **An outside instrument** Use another exogenous variable that predicts the regressor without correlating with $u - \beta_1 e_1$.

Two measures, one true signal

Separate reporting noise needs its own argument

First report, then second report

$x_1 = x_1^* + e_1, \qquad z_1 = x_1^* + a_1$



Shared signal motivates relevance; the noise restrictions determine exogeneity.

Three conditions for a second measure

Signal, outcome-error exogeneity and separate noise

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + (u - \beta_1 e_1).$$

- **Same signal** $z_1 = x_1^* + a_1$: both reports share the true variable; maintain classical noise restrictions.
- **Original outcome error** $\text{Cov}(z_1, u) = 0$: the second report avoids the other omitted determinants of y.
- **First report's noise** $\text{Cov}(a_1, e_1) = 0$: when one report overstates, the other does not systematically overstate too.

Why relevance holds | Why exogeneity holds

Example of unrelated measurement errors

A concrete reason to examine the separate-noise assumption

Same true annual salary

$x_1 = x_1^* + e_1$ is the worker report; $z_1 = x_1^* + a_1$ is the employer report.

- **Possible separate errors**
  - A worker forgets an irregular bonus when recalling annual pay;
  - An employer assigns a payment to the wrong worker because of a payroll matching error.
- These mechanisms need not move together.

When a second report is unavailable, consider the other remedy: an outside instrument.

An outside instrument must avoid the measurement noise

The second remedy does not require a second report

Observed schooling equation

$\log(wage) = \beta_0 + \beta_1 educ^{obs} + \mathbf{c}'\boldsymbol{\gamma} + (u - \beta_1 e).$

$\mathbf{c}$ contains the included exogenous variables; $educ^{obs} = educ^{*} + e.$ An outside variable such as parental education is a candidate if it:

- predicts schooling after the included exogenous variables - relevance
- is uncorrelated with both omitted wage determinants u and schooling-report noise e. - exogenous

Then it avoids the composite error $u - \beta_1 e$. Family-background association with u remains a reason to question this candidate.

Understand the measurement process before choosing a remedy

Where the error enters determines the bias and the useful extra information

- **Locate it** Is the noisy variable the outcome or a regressor? Is the error related to the regressors or true signal?
- **Explain it** Recall mistakes, recording mistakes and shared omissions imply different error correlations and potentially different bias directions.
- **Find information** A second measure or outside instrument helps only if its noise and outcome-error relations satisfy the needed IV conditions.

Numerical attenuation example | Optional ability indicators

Contents

Intro 15-5a

1. Two Stage Least Squares
2. IV Solutions to Errors-in-Variables Problems
3. **Testing for Endogeneity and Overidentifying Restrictions**
   - **Testing for Endogeneity**
   - Testing Overidentifying Restrictions
4. 2SLS with Heteroskedasticity

Technical extensions are linked in the appendix; Midterm Review is a separate deck.

IV can solve endogeneity problems, but costs precision. Do we need it for this regressor?

Why test for endogeneity?

OLS is more efficient when its explanatory variables are exogenous

Same structural target

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + \beta_3 z_2 + u_1$.

- Under exogeneity and homoskedasticity, OLS is consistent and more efficient.
- IV uses instrument-predicted variation and can have much larger standard errors.
- Is the regressor endogenous, making the IV correction necessary?

Under endogeneity, OLS and valid IV generally have different probability limits.

Recall OLS and IV precision | The Hausman comparison

Testing for endogeneity

Compare OLS and IV under maintained instrument exogeneity

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + \beta_3 z_2 + u_1. \tag{15.49}$$

Additional excluded IVs: z_3, z_4. Maintain IV exogeneity and the rank condition.

$$H_0 : \text{Cov}(y_2, u_1) = 0.$$

Under the null, both OLS and 2SLS are consistent. A statistically meaningful difference can provide evidence of endogeneity.

Locate the potentially endogenous component

Equation 15.50 separates prediction from the remaining variation

Structural model

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + \beta_3 z_2 + u_1$; all z_j are maintained exogenous.

$$y_2 = \underbrace{\pi_0 + \pi_1 z_1 + \pi_2 z_2 + \pi_3 z_3 + \pi_4 z_4}_{y_2^*} + v_2. \tag{15.50}$$

$$\text{Cov}(y_2, u_1) = \text{Cov}(y_2^*, u_1) + \text{Cov}(v_2, u_1) = \text{Cov}(v_2, u_1).$$

The first term is zero because each z_j is uncorrelated with u_1. Any remaining regressor–error association must come through v_2.

Delta measures the endogenous association

Project the structural error on the first-stage residual

From the first stage

$\mathrm{Cov}(y_2, u_1) = \mathrm{Cov}(v_2, u_1)$, because the predicted part uses valid instruments.

$$u_1 = \delta_1 v_2 + e_1, \qquad \delta_1 = \frac{\mathrm{Cov}(v_2, u_1)}{\mathrm{Var}(v_2)}.$$

This defines e_1, the remainder orthogonal to v_2. Therefore

$$\mathrm{Cov}(y_2, u_1) = \mathrm{Cov}(v_2, u_1) = \delta_1 \mathrm{Var}(v_2).$$

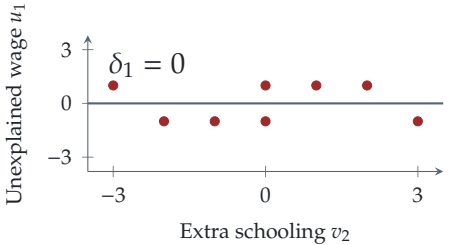
With positive residual variance, $\delta_1 = 0$ is exactly the regressor-exogeneity condition.

Schooling and unexplained wages

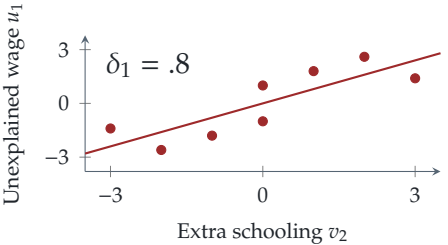
15-5a: the null is that schooling is exogenous

Null $H_0 : \delta_1 = 0$: schooling is exogenous.

v_2: extra schooling; u_1: other wage factors, such as ability.



Exogenous schooling: its leftover part is also unrelated to other wage factors.



Ability can raise both schooling and wages, creating the upward link.

Test whether schooling is endogenous

15-5a: does the extra-schooling component predict wages?

Step 1 Predict schooling from both parents and $\mathbf{c} = (exper, exper^2)'$; save $\hat{v}_2 = educ - \widehat{educ}$.

Step 2 Add $\hat{v}_2$ to the wage regression:

$$\log(wage) = \beta_0 + \beta_1 educ + \boldsymbol{\gamma}'\mathbf{c} + \delta_1 \hat{v}_2 + error. \tag{15.51}$$

Null $H_0 : \delta_1 = 0$: schooling is exogenous. Reject for a significant extra-schooling coefficient.

MROZ (conventional test) $\hat{\delta}_1 = .058$, $t = 1.67$, $p \simeq .095$. At 5%, do not reject: insufficient evidence of schooling endogeneity.

2SLS coefficient equality | Worked schooling test | Several regressors

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Intro 15-5b

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   - Testing for Endogeneity
   - **Testing Overidentifying Restrictions**
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Technical extensions are linked in the appendix; Midterm Review is a separate deck.

We maintained valid instruments. Can extra instruments help us examine that assumption?

One excluded instrument identifies this equation

Keep the same included variables and structural effect

Same structural model

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + \beta_3 z_2 + u_1$.

- The included exogenous variables z_1, z_2 act as their own instruments.
- Use z_3 as the excluded IV for y_2: maintain $\text{Cov}(z_3, u_1) = 0$ and relevance after z_1, z_2.
- One excluded IV for one endogenous regressor: the equation is **just identified**.

A second instrument adds a restriction, not a parameter

The same equation now has one overidentifying restriction

Same structural model

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + \beta_3 z_2 + u_1$.

$\text{Cov}(z_3, u_1) = \text{Cov}(z_4, u_1) = 0$; each IV predicts y_2 beyond z_1, z_2.

Estimate the same β_1 using z_3 alone or z_4 alone.

Two nonredundant excluded IVs, one endogenous regressor: **one overidentifying restriction**.

Valid IVs target one slope. Are their estimates compatible?

Two parents provide two estimates of the same effect

Example 15.5: keep the wage model, experience controls and sample fixed

$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u_1$. Same 428 women. Instrument schooling with each parent separately:

Instrument choice	Schooling slope	SE (rounded)
Mother only	.0493	.037
Father only	.0702	.034

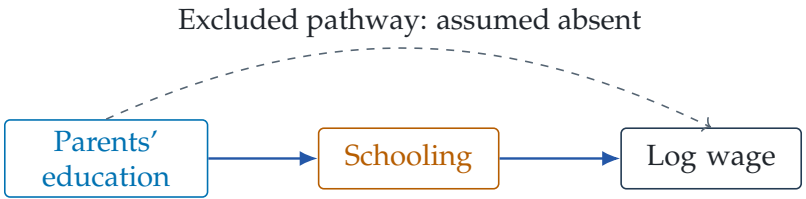
Valid parents target the same slope. Are the differences more than sampling variation?

Can both instruments fit one wage effect?

15-5b: extra restrictions let us examine what remains unexplained

Same model

$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u_1$.



Under valid IV restrictions, the parents should not predict the error left after removing schooling and experience effects. Check the leftovers next.

Can the instruments predict the structural residual?

15-5b: use actual schooling, then regress the leftovers on all exogenous variables

First estimate 2SLS Use both parents, retaining experience and its square. **Save the structural residual**

$$\hat{u}_{i1} = \log(wage_i) - \hat{\beta}_0 - \hat{\beta}_1 educ_i - \hat{\beta}_2 exper_i - \hat{\beta}_3 exper_i^2.$$

Then ask what predicts it

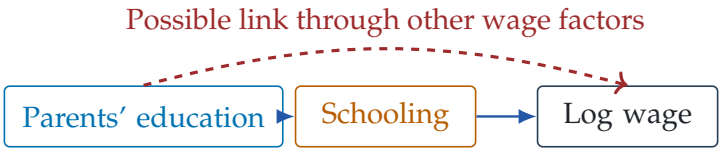
$$\hat{u}_{i1} = a_0 + a_1 exper_i + a_2 exper_i^2 + a_3 motheduc_i + a_4 fatheduc_i + e_i.$$

The residual regression's R_1^2 measures remaining fit. This is not the first-stage residual $\hat{v}_2$.

Can parents still explain the leftover wages?

15-5b: look for a wage link beyond schooling and experience

Null Parents' education is unrelated to the other factors in u_1.



Large residual R_1^2: parents still explain wages after removing schooling and experience effects.

Unusually large nR_1^2: reject the IV restrictions.

$nR_1^2 \overset{a}{\sim} \chi_1^2$. Assumes homoskedasticity and strong identification.

Limits and exact identification | Which diagnostic answers which question?

The parents pass the residual-fit check

Example 15.8: little remaining wage prediction by the parents

Same residual regression
$\hat{u}_1$ on 1, *exper*, $exper^2$, *motheduc*, *fatheduc*.

n	R_1^2	nR_1^2	p-value
428	.0008833	**.3781**	**.539**

Do not reject H_0 at 5%: parents explain very little of the leftover wage variation.

Compare the three diagnostic questions

Contents

Intro 15-6

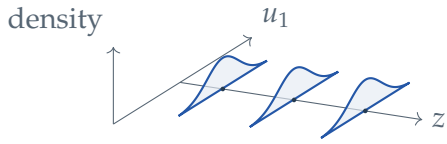
1. Two Stage Least Squares
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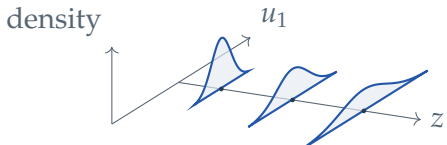
Even with valid instruments, what if the structural error spread varies across observations?

Heteroskedasticity and IV inference

2 solutions



constant spread



changing spread

Illustration only: $E(u_1 \mid z) = 0$; the variance can vary.

Robust SEs: keep the 2SLS coefficients; recompute uncertainty.

Reweighting: changes estimation under extra variance and exogeneity conditions.

Robust output and interval | Weighting intuition

Key takeaways from instrumental variables

One target, a credible source of variation, and honest uncertainty

- **Identify the effect** Exogeneity rules out the error channel; relevance supplies separate regressor variation. *15-1–15-2.*
- **Estimate it** 2SLS combines valid instruments in a first-stage prediction. Strength and collinearity determine precision. *15-3.*
- **Diagnose the problem** Measurement mechanisms guide IV remedies. Residual tests probe regressor endogeneity; extra-IV comparisons probe instrument compatibility. *15-4–15-5.*
- **Report the evidence** Interpret the effect and uncertainty together. Explain the economic argument for instrument exogeneity. *15-6.*

Appendix contents

Next Lecture

Prepare for the midterm and Problem Set 2

- Today: Instrumental Variables
  Introductory Econometrics, 15-3–15-6
  Advanced reference: Wooldridge (2010), Chapter 5
- Next: Midterm Exam (30%), Monday 26 October 2026, 09:00–11:00 — covers L1–L4
- Then: Additional Single-Equation Topics
  Wooldridge (2010), Chapter 6 (Week 9)
- Problem Set 2 due Friday 23 October 2026

Exams are closed book and in person; one double-sided A4 cheat sheet is allowed.

Adapted from slides by Shuyang Sheng and Qihui Chen

Appendix contents

Choose an explanation, diagnostic or research application

Estimation and uncertainty

Assumptions and large-sample results

Best linear prediction

One-IV and two-stage equality

Residuals and IV standard errors

Stata, R and Python commands

Diagnostics and applications

Midterm venue map

Measurement error and tests

Midterm Review is a separate slide deck.

Research applications

Back to lecture takeaways

Appendix contents: measurement and tests

Detailed arguments stay behind the main intuition

Partialling-out algebra

Attenuation derivation

Second-measure relevance

Second-measure exogeneity

Several endogenous regressors

Coefficient tests after 2SLS

Residual-augmentation algebra

Overidentification limits

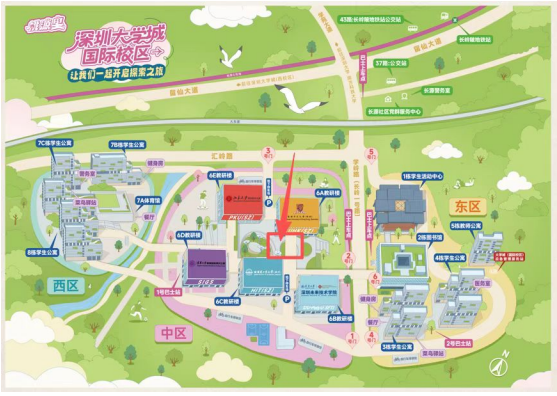
Testing changing error variance

Weighted 2SLS

Back to appendix contents

Midterm venue map

西丽小礼堂


深圳大学城
国际校区
让两个一起开启精彩之旅
留仙大道
学苑大道
西丽湖
汇枢路
7C南学生公寓
7B南学生公寓
7A南学生公寓
8南学生公寓
7南食堂
7南超市
7南母婴室
7南教育超市
7A南育馆
7南银行
7南医务室
7南理发店
7南邮政
7南快递
7南图书馆
7南教师公寓
7南学生活动中心
7南教师发展中心
7南学生公寓
7南食堂
7南超市
7南银行
7南母婴室
7南理发店
7南快递
3南学生公寓
2南巴士站
1南巴士站
6A教研楼
6B教研楼
6C教研楼
6D教研楼
5A教研楼
5B教研楼
5C教研楼
5D教研楼
5E教研楼
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5T教研楼
5U教研楼
5V教研楼
5W教研楼
5X教研楼
5Y教研楼
5Z教研楼
深圳大学生命与海洋科学学院
深圳大学生物医学工程学院
深圳大学医学部
深圳大学医学院
深圳大学生物医学工程学院
深圳大学医学部
深圳大学医学院
西区
中区
东区
47路长树隧地铁站公交站
长树隧地铁站
37路公交站
长源置地港
长源置地港
长源置地港

More noise leaves a smaller signal share

L3b simple-model recap: the OLS probability limit moves toward zero

Same model

$y = \beta_0 + \beta_1 x_1^* + u$, $x_1 = x_1^* + e_1$; maintain classical noise and true-regressor exogeneity.

$$\operatorname{plim} \hat{\beta}_{1,OLS} = \beta_1 \frac{\operatorname{Var}(x_1^*)}{\operatorname{Var}(x_1^*) + \operatorname{Var}(e_1)}.$$

The signal share lies between zero and one. This is a probability limit, not an exact finite-sample bias formula.

Attenuation after included variables | Back to attenuation intuition

Shared signal supplies partial relevance

Remove included-variable variation before checking the second report

Two reports

$x_1 = x_1^* + e_1, \qquad z_1 = x_1^* + a_1.$

Assume $\text{Cov}(a_1, e_1) = 0$, $\text{Cov}(e_1, x_1^*) = \text{Cov}(e_1, x_2) = 0$, and $\text{Cov}(a_1, x_1^*) = \text{Cov}(a_1, x_2) = 0$. Tildes remove population linear predictions on $1, x_2$:

$$\text{Cov}(\widetilde{z}_1, \widetilde{x}_1) = \text{Var}(\widetilde{x}_1^*) > 0.$$

The shared true signal must retain positive variance after x_2.

Back to second-measure conditions

Separate noise supplies instrument exogeneity

Check the second report against the entire transformed error

Outcome error

$u - \beta_1 e_1$; instrument $z_1 = x_1^* + a_1$.

$$\operatorname{Cov}(z_1, u - \beta_1 e_1) = \operatorname{Cov}(z_1, u) - \beta_1 \operatorname{Cov}(z_1, e_1).$$

The first term is zero by assumption; the second is zero because

$$\operatorname{Cov}(z_1, e_1) = \operatorname{Cov}(x_1^*, e_1) + \operatorname{Cov}(a_1, e_1) = 0.$$

Therefore $\operatorname{Cov}(z_1, u - \beta_1 e_1) = 0$.

Back to second-measure conditions

Assumptions 2SLS.1–2SLS.4

Intro 15A states the conditions for consistency

- **2SLS.1: Linear in parameters**
  $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$.
- **2SLS.2: Random sampling** A random sample on y, the x_j and the z_j.
- **2SLS.3: Rank** No perfect linear relationships among IVs; the rank condition for identification holds.
- **2SLS.4: Exogenous IVs**
  $E(u) = 0, \quad \text{Cov}(z_j, u) = 0$ for every IV.

Theorems and homoskedasticity

Back to the 2SLS conditions

Intro 15A: consistency and normal inference

Homoskedasticity is an additional condition

Theorem 15A.1 Under Assumptions 2SLS.1–2SLS.4, the 2SLS estimator is consistent.

2SLS.5: Homoskedasticity For the collection of IVs $\mathbf{z}$,

$$E(u^2 \mid \mathbf{z}) = \sigma^2.$$

Theorem 15A.2 Under Assumptions 2SLS.1–2SLS.5, the 2SLS estimators are asymptotically normally distributed.

Estimate the variance using the structural residuals and the IV variance formula. With heteroskedasticity, adjust the covariance.

Back to the 2SLS conditions

Linear combinations preserve exogeneity

Relevance is a separate requirement

Maintained moments
$E(u_1) = 0, \text{Cov}(z_j, u_1) = 0$ for $j = 1, 2, 3$.

For fixed coefficients a_j, let $a = a_0 + \sum_{j=1}^{3} a_j z_j$.

$$\text{Cov}(a, u_1) = \sum_{j=1}^{3} a_j \, \text{Cov}(z_j, u_1) = 0.$$

This establishes orthogonality. A particular linear combination can still have zero predictive power through cancellation.

Back to combining the instruments

Why the projection maximizes correlation

Projection orthogonality and Cauchy–Schwarz give the result

$y_2 = y_2^* + v_2$, with $\text{Cov}(z_j, v_2) = 0$. Let a be a nonconstant linear combination of the IVs.

$$\text{Cov}(y_2, a) = \text{Cov}(y_2^*, a).$$

$$\text{Corr}(y_2, a)^2 \leq \frac{\text{Var}(y_2^*)\,\text{Var}(a)}{\text{Var}(y_2)\,\text{Var}(a)} = \frac{\text{Var}(y_2^*)}{\text{Var}(y_2)}.$$

Equality holds for $a = y_2^*$ when its variance is positive. Other combinations can be irrelevant despite being exogenous.

Back to the population prediction

The projection minimizes squared prediction error

Population coefficients are pi; candidate coefficients are a

For candidate coefficients $a = (a_0, a_1, a_2, a_3)$, the population projection solves

$$\pi = \arg\min_{a} E[(y_2 - a_0 - a_1 z_1 - a_2 z_2 - a_3 z_3)^2].$$

The first-order conditions give $E(v_2) = 0$ and $E(z_j v_2) = 0$, where $v_2 = y_2 - \pi_0 - \pi_1 z_1 - \pi_2 z_2 - \pi_3 z_3$.

The minimizing coefficients are π_j; hats denote their sample estimates.

The correlation bound | Back to the population prediction

Why excluded relevance identifies the scalar effect

Included-variable moments alone leave one slope undetermined

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1.$$

$$E(u_1) = 0, \qquad E(z_1 u_1) = 0.$$

Two moments cannot generally determine $\beta_0, \beta_1, \beta_2$. An excluded instrument must supply information beyond z_1.

The remaining scalar moment

Back to excluded predictive variation

Use an excluded IV after removing the included variable

A nonzero denominator determines the remaining slope

Residualized equation

$\tilde{y}_1 = \beta_1 \tilde{y}_2 + u_1$. Tildes remove population predictions on $1, z_1$; z_j, $j \in \{2,3\}$, is an excluded IV.

$$E(\tilde{z}_j u_1) = 0 \quad \Longrightarrow \quad \beta_1 = \frac{E(\tilde{z}_j \tilde{y}_1)}{E(\tilde{z}_j \tilde{y}_2)}, \qquad E(\tilde{z}_j \tilde{y}_2) \neq 0.$$

Relevance requires a nonzero denominator.

Back to excluded predictive variation

Population projection and sample OLS

Both solve least-squares prediction problems

Let $\mathbf{z}_i = (1, z_{i1}, z_{i2}, z_{i3})'$. **Population** Minimize expected squared error:

$$\pi = \arg\min_a E[(y_2 - \mathbf{z}'a)^2].$$

Sample Minimize the sample average:

$$\hat{\pi} = \arg\min_a \frac{1}{n} \sum_i (y_{i2} - \mathbf{z}_i'a)^2.$$

This is OLS; multiplying by n leaves the minimizer unchanged.

Compare their normal equations | Back to population and sample

Matching normal equations

Both concern prediction residuals, not the structural error

Prediction residual	Orthogonality
$v_2 = y_2 - \mathbf{z}'\pi$	$E(\mathbf{z}v_2) = 0$
$\hat{v}_{i2} = y_{i2} - \mathbf{z}_i'\hat{\pi}$	$\sum_i \mathbf{z}_i\hat{v}_{i2} = 0$

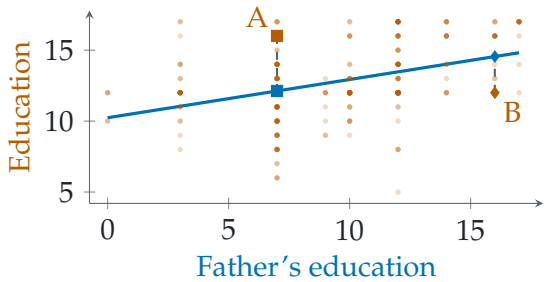
IV exogeneity is a separate maintained assumption: $E(\mathbf{z}u_1) = 0$.

Back to population and sample

First stage: predict schooling

One-IV recap: the father-only model just introduced

$$educ = \pi_0 + \pi_1 fatheduc + r$$



Each dot is one woman.

Keep the fitted schooling $\widehat{educ}_i$.

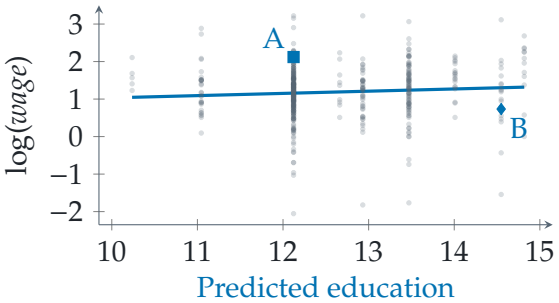
Separate the remainder $educ_i - \widehat{educ}_i$. Instrument exogeneity is the maintained assumption about the fitted part.

Second-stage picture | Back to one-IV equivalence

Second stage: use predicted schooling

Example 15.1: retain the instrument-linked component

$$\log(wage)_i = a + b\widehat{educ}_i + \eta_i$$



Same women A and B
Use fitted schooling on the horizontal axis; log wages stay fixed.

IV slope .0592.

Back to one-IV equivalence

A fitted value is predicted schooling, not a wage

Example 15.5: calculate one woman's first-stage prediction

First woman
exper = 14, *motheduc* = 12, *fatheduc* = 7.

$$\widehat{educ} = 9.1026 + .0452 exper - .0010 exper^2 + .1576 motheduc + .1895 fatheduc.$$

What fitted education does this rule assign to her?

$$9.1026 + .0452(14) - .0010(14^2) + .1576(12) + .1895(7) \simeq 12.76.$$

Use her fitted value with her wage and controls in stage two.

Back to the first-stage prediction

First stage: predict the regressor with one IV

The same sample and an intercept in both regressions

$y_i = \beta_0 + \beta_1 x_i + u_i$; one IV: z_i.

First-stage OLS

$$\hat{x}_i = \hat{\pi}_0 + \hat{\pi}_1 z_i, \qquad \hat{\pi}_1 = \frac{\widehat{\mathrm{Cov}}(z, x)}{\widehat{\mathrm{Var}}(z)}.$$

The direct IV formula is

$$\hat{\beta}_{1,IV} = \frac{\widehat{\mathrm{Cov}}(z, y)}{\widehat{\mathrm{Cov}}(z, x)}.$$

Sample moments: $\widehat{\mathrm{Var}}(z) > 0$ and $\widehat{\mathrm{Cov}}(z, x) \neq 0$.

Substitute the fitted value | Back to one-IV equivalence

Second stage: substitute the fitted value

Constants drop from sample covariance and variance

First-stage fit: $\hat{x}_i = \hat{\pi}_0 + \hat{\pi}_1 z_i$. OLS of y_i on $\hat{x}_i$, with an intercept, gives

$$\hat{\beta}_{1,2SLS} = \frac{\widehat{\text{Cov}}(\hat{x}, y)}{\widehat{\text{Var}}(\hat{x})}.$$

Substitute the first-stage fit:

$$\widehat{\text{Cov}}(\hat{x}, y) = \hat{\pi}_1 \widehat{\text{Cov}}(z, y),$$
$$\widehat{\text{Var}}(\hat{x}) = \hat{\pi}_1^2 \widehat{\text{Var}}(z).$$

Cancel the first-stage slope | Back to one-IV equivalence

Cancel the first-stage slope

One excluded IV makes the coefficient formulas identical

Use $\hat{\pi}_1 = \widehat{\text{Cov}}(z,x)/\widehat{\text{Var}}(z) \neq 0$.

$$\hat{\beta}_{1,2SLS} = \frac{\hat{\pi}_1 \widehat{\text{Cov}}(z,y)}{\hat{\pi}_1^2 \widehat{\text{Var}}(z)} = \frac{\widehat{\text{Cov}}(z,y)}{\hat{\pi}_1 \widehat{\text{Var}}(z)}.$$

Substitute $\hat{\pi}_1$ and cancel the sample variance:

$$\hat{\beta}_{1,2SLS} = \frac{\widehat{\text{Cov}}(z,y)}{[\widehat{\text{Cov}}(z,x)/\widehat{\text{Var}}(z)]\widehat{\text{Var}}(z)} = \frac{\widehat{\text{Cov}}(z,y)}{\widehat{\text{Cov}}(z,x)} = \hat{\beta}_{1,IV}.$$

First-stage OLS gives $\overline{\hat{x}} = \bar{x}$, so the intercepts also agree. Coefficients agree; inference still needs IV standard errors.

Back to one-IV equivalence

The residuals differ at every candidate coefficient

Actual schooling and predicted schooling define different regression errors

Define $\hat{v}_{i2} = y_{i2} - \widehat{y}_{i2}$. At any $b = (b_0, b_1, b_2)'$:

$$u_{i1}(b) = y_{i1} - b_0 - b_1 y_{i2} - b_2 z_{i1},$$
$$e_{i1}(b) = y_{i1} - b_0 - b_1 \widehat{y}_{i2} - b_2 z_{i1}.$$

Subtract:

$$u_{i1}(b) = e_{i1}(b) - b_1 \hat{v}_{i2}.$$

Why the moment equations coincide | Back to IV and OLS equivalence

First-stage orthogonality removes the difference

The instruments and second-stage regressors are the same vector

For $\mathbf{w}_i = (1, \widehat{y}_{i2}, z_{i1})'$, first-stage OLS gives $\sum_i \mathbf{w}_i \hat{v}_{i2} = 0$: each element is a first-stage regressor or their linear combination. Weight the residual identity by $\mathbf{w}_i$ and sum:

$$\sum_i \mathbf{w}_i u_{i1}(b) = \sum_i \mathbf{w}_i e_{i1}(b).$$

SSR normal equations | Back to coefficient equality

Second-stage least squares solves the IV equations

Full rank gives the same unique coefficients

IV definition Solve $\sum_i \mathbf{w}_i u_{i1}(b) = 0$. **Second-stage OLS** Minimize $\sum_i e_{i1}(b)^2$. Its normal equations are

$$\frac{\partial}{\partial b} \sum_i e_{i1}(b)^2 = -2 \sum_i \mathbf{w}_i e_{i1}(b) = 0.$$

Full rank gives the same unique coefficients on the same sample and controls.

The minimized SSR uses e, not structural u. Validity remains a separate assumption.

Back to IV and OLS equivalence

Population and sample IV moments

The structural residual is the same in all three sample conditions

Structural residual

$\hat{u}_{i1} = y_{i1} - \hat{\beta}_0 - \hat{\beta}_1 y_{i2} - \hat{\beta}_2 z_{i1}$.

Population condition	Sample analogue
$E(u_1) = 0$	$\sum_i \hat{u}_{i1} = 0$
$E(z_1 u_1) = 0$	$\sum_i z_{i1} \hat{u}_{i1} = 0$
$E(y_2^* u_1) = 0$	$\sum_i \widehat{y}_{i2} \hat{u}_{i1} = 0$

These three sample equations jointly determine $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$.

Back to IV equations

Why the composite error is orthogonal

IV exogeneity is assumed; projection orthogonality follows from least squares

Composite error

$\tilde{u}_1 = u_1 + \beta_1 v_2$.

Two sources of zero covariance

- IV exogeneity: $\text{Cov}(y_2^*, u_1) = 0$, since y_2^* is a linear combination of valid IVs.
- Projection orthogonality: $\text{Cov}(y_2^*, v_2) = 0$.

$$\begin{aligned}\text{Cov}(y_2^*, \tilde{u}_1) &= \text{Cov}(y_2^*, u_1) + \beta_1 \,\text{Cov}(y_2^*, v_2) \\ &= 0 + \beta_1(0) = 0.\end{aligned}$$

Back to the substitution intuition

Compute structural residuals

Use actual schooling in the wage equation

$$\hat{u}_{i1} = \log(wage_i) - \hat{\beta}_0 - \hat{\beta}_1 educ_i - \hat{\beta}_2 exper_i - \hat{\beta}_3 exper_i^2.$$

Schooling standard error	Value
IV: structural residuals	.03144
Naive second-stage OLS	.03296

IV versus naive SEs | Two residuals | Variance calculation

Back to IV software and uncertainty

Structural and second-stage residuals

The difference is the fitted coefficient times the first-stage residual

$$\hat{u}_{i1} = \log(wage_i) - \hat{\beta}_0 - \hat{\beta}_1 educ_i - \hat{\beta}_2 exper_i - \hat{\beta}_3 exper_i^2.$$

$$\hat{e}_{i1} = \log(wage_i) - \hat{\beta}_0 - \hat{\beta}_1 \widehat{educ}_i - \hat{\beta}_2 exper_i - \hat{\beta}_3 exper_i^2.$$

$$\hat{e}_{i1} - \hat{u}_{i1} = \hat{\beta}_1 (educ_i - \widehat{educ}_i).$$

Back to structural residuals

Same woman, two different residuals

Actual schooling gives the structural error estimate

$$\log(wage_i) = \hat{\beta}_0 + \hat{\beta}_1 educ_i + \mathbf{c}_i'\hat{\gamma} + \hat{u}_i.$$

For this woman, $educ = 12$, $\widehat{educ} = 12.7560$, $\log(wage) = 1.2102$.

Schooling used	Predicted log wage	Residual
Actual: 12	1.2270	−.0169
Fitted: 12.7560	1.2735	−.0633

Back to structural residuals

Estimating structural error variance

Use the number of structural coefficients in the degrees of freedom

Two-parent MROZ

$n = 428$; intercept plus 3 slopes, so $k + 1 = 4$.

$$\hat{\sigma}^2 = \frac{\sum_i \hat{u}_{i1}{}^2}{n - k - 1} = \frac{193.0200}{424} = .45524.$$

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2 (\widehat{X}' \widehat{X})^{-1}.$$

Columns of $\widehat{X}$: $1, \widehat{educ}, exper, exper^2$. This conventional covariance uses Intro 2SLS.1–2SLS.5.

Back to structural residuals

Estimate 2SLS in Stata

Specify the outcome, explanatory variables and excluded instruments

Same MROZ model
Outcome: *lwage*; endogenous: *educ*; included exogenous: *exper*, *expersq*; excluded IVs: *motheduc*, *fatheduc*.

```
ivregress 2sls lwage exper expersq ///
(educ = motheduc fatheduc), small
```

The small option matches the conventional degrees-of-freedom correction shown here.

R command | Python command | Back to IV software and uncertainty

Estimate 2SLS in R

Specify the outcome, explanatory variables and excluded instruments

Same MROZ model
Outcome: *lwage*; endogenous: *educ*; included exogenous: *exper*, *expersq*; excluded IVs: *motheduc*, *fatheduc*.

```
ivreg(lwage ~ educ + exper + expersq |
  exper + expersq + motheduc + fatheduc,
  data = d)
```

Use the `ivreg` package. Its conventional covariance matches the comparison shown here.

Back to IV software and uncertainty

Estimate 2SLS in Python

Specify the outcome, explanatory variables and excluded instruments

Same MROZ model

lwage on *educ*, *exper*, *expersq*; excluded IVs: *motheduc*, *fatheduc*.

```
IV2SLS.from_formula(
  "lwage ~ 1 + exper + expersq + "
  "[educ ~ motheduc + fatheduc]",
  data=d).fit(cov_type="unadjusted",
  debiased=True)
```

`linearmodels.iv`; options match the conventional covariance.

Back to IV software and uncertainty

The partialling-out algebra

Residual regressor variation determines a separate slope

OLS model

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1$.

For observation i, let r_{i2} be the sample OLS residual from regressing y_2 on 1, z_1; let r_{i1} be the analogous outcome residual.

$$\hat{\beta}_1 = \frac{\sum_i r_{i2} r_{i1}}{\sum_i r_{i2}^2}, \qquad \sum_i r_{i2}^2 = SST_2(1 - R_2^2).$$

With fitted $\widehat{y_2}$, repeat the same removal of 1, z_1 to obtain the IV denominator in equation15.43.

Back to the remaining-variation picture

Partialling out reproduces the OLS slope

The partialled regression reproduces the full OLS slope

Full OLS

$colGPA_i = \beta_0 + \beta_1 hsGPA_i + \beta_2 ACT_i + u_i.$

1. Regress $hsGPA$ on an intercept and ACT. Keep $\hat{r}_{i1} = hsGPA_i - \widehat{hsGPA}_i$.
2. Regress $colGPA$ on an intercept and $\hat{r}_{i1}$; call its slope $\hat{b}_1$.

$$\hat{b}_1 = \hat{\beta}_1^{\text{full OLS}}.$$

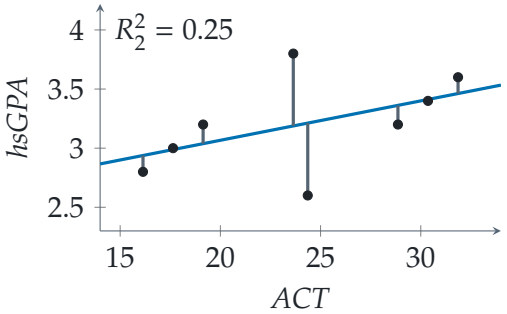
$\hat{r}_{i1}$ isolates GPA variation beyond ACT. This **FWL identity** concerns the coefficient, not causality.

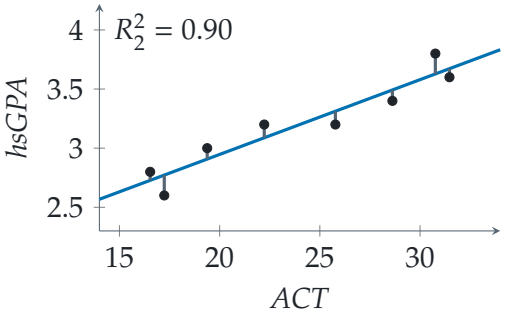
GPA residual gaps | Back to OLS precision

More overlap leaves smaller residual gaps

College GPA on high-school GPA and ACT: compare residual gaps

$colGPA = \beta_0 + \beta_1 hsGPA + \beta_2 ACT + u, \quad SST_2 = 1.12.$

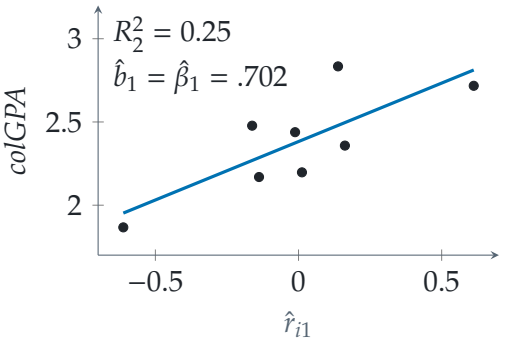


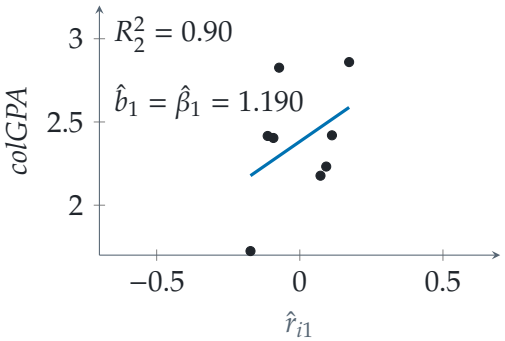


Residual-regression plots | Back to OLS precision

Smaller gaps leave less horizontal spread

Less unique spread reduces precision; the two samples need not have equal slopes





Partialling-out algebra | Back to the precision benchmark

More than one endogenous regressor

15-3d: optional; use every instrument in every first stage

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 y_3 + \beta_3 z_1 + \beta_4 z_2 + \beta_5 z_3 + u_1. \tag{15.44}$$

Suppose z_4, z_5 are additional exogenous variables excluded from this model.

- Regress each of y_2, y_3 on all five IVs and an intercept.
- In stage two retain z_1, z_2, z_3 and replace both endogenous regressors by their fitted values.
- Two endogenous regressors require at least two excluded IVs; this count is necessary, not sufficient.

Why counts alone are insufficient | Back to first-stage prediction

Order and rank

Two instruments can supply only one direction of variation

After controls

$y = \beta_1 x_1 + \beta_2 x_2 + u$, with exogenous instruments.

Population reduced-form slopes after controls:

Excluded IV	Slope for x_1	Slope for x_2
z_1	1	2
z_2	2	4

Both identify only $\beta_1 + 2\beta_2$. The second direction repeats the first.

Alternative slope pairs | Rank condition | Back to first-stage prediction

Identification requires unique coefficients

Different slope pairs can fit the same population moments

Same first-stage directions

$z_1 : (1, 2)$, $z_2 : (2, 4)$. Outcome responses are 3 and 6.

Candidate slopes	$\beta_1 + 2\beta_2$	$2\beta_1 + 4\beta_2$
$(1, 1)$	3	6
$(3, 0)$	3	6

Both fit the moments: the two slopes are not separately identified.

Back to first-stage prediction

Rank and population moments

The matrix statement supports Intro Assumption 2SLS.3

Column vectors

$y = \mathbf{x}'\beta + u$. Here $\mathbf{x}$ is $(k+1)\times 1$ and $\mathbf{z}$ is $L\times 1$; both include the intercept, and $\mathbf{z}$ includes all exogenous regressors.

$$E(\mathbf{z}u) = 0 \quad \Longrightarrow \quad E(\mathbf{z}\mathbf{x}')\beta = E(\mathbf{z}y).$$

$$\operatorname{rank} E(\mathbf{z}\mathbf{z}') = L, \qquad \operatorname{rank} E(\mathbf{z}\mathbf{x}') = k+1.$$

The second condition gives enough independent moment restrictions to determine the coefficient vector; $L \geq k+1$ alone is insufficient.

Practical rank checks | Back to independent directions

Practical checks of the rank condition

Check relevance for the whole instrument–regressor system

Identification requirement

rank $E(\mathbf{z}\mathbf{x}') = k + 1$, with all included exogenous regressors in $\mathbf{z}$.

One endogenous variable

Run its full first stage and jointly test only the excluded-IV coefficients. A conventional F uses first-stage homoskedasticity; a robust Wald test allows it to fail.

Rejection supplies evidence of nonzero partial relevance. Strong identification and instrument exogeneity require further assessment.

Several endogenous variables | Back to independent directions

Rank checks with several regressors

Significant separate regressions can still repeat one direction

Same requirement
rank $E(\mathbf{z}\mathbf{x}') = k + 1$.

Use the same complete IV list in every first stage.

Full rank requires independent prediction directions across the regressors.

Sample moment matrix: $Z'X/n$.

Check these directions jointly in the system.

Back to independent directions

Test a coefficient with its IV standard error

15-3e: optional; use the large-sample IV reference

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u_1.$$

$$H_0 : \beta_1 = b_{1,0}, \qquad t = \frac{\hat{\beta}_1 - b_{1,0}}{\text{se}_{IV}(\hat{\beta}_1)} \overset{a}{\sim} N(0,1).$$

Null	Estimate	IV SE	t
$\beta_1 = 0$	.06140	.03144	1.953

This does not reach the approximate two-sided 5% cutoff 1.96.

Joint coefficient test | Back to the first-stage block

A joint test asks a joint economic question

Use the covariance from the IV estimate

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u_1.$$

$$H_0 : \beta_2 = \beta_3 = 0.$$

Do the experience terms matter jointly, conditional on schooling?

Conventional IV test	Degrees of freedom	p
Wald $\chi^2 = 19.64$	2	.000054

The usual OLS SSR-difference formula is not a valid shortcut after 2SLS.

Why the IV fit shortcut fails | Back to first-stage prediction

Why IV R-squared is not an OLS fit measure

Use IV covariance for joint tests, not an OLS SSR shortcut

$$\hat{u}_i = y_i - \mathbf{x}_i'\hat{\beta}_{IV}, \qquad R^2_{IV} = 1 - \frac{\sum_i \hat{u}_i{}^2}{\sum_i (y_i - \bar{y})^2}.$$

- IV does not minimize the outcome equation's residual sum of squares. Its SSR can exceed SST, making this R^2 negative.
- Restricted and unrestricted IV fits need not have the OLS SSR ordering.
- Use the covariance-based joint test reported by IV software. Fit measures prediction; validity rests on the IV assumptions.

Back to joint coefficient tests

The Hausman comparison

Evaluate the coefficient difference against its sampling uncertainty

Null
The regressor is exogenous; the instruments are maintained valid.

Under the efficient-OLS assumptions, compare β_1. Let $\widehat{V}_{1,j}$ estimate $\text{Var}(\hat{\beta}_{1,j})$, for $j = IV, OLS$.

$$d = \hat{\beta}_{1,IV} - \hat{\beta}_{1,OLS}, \qquad H = \frac{d^2}{\widehat{V}_{1,IV} - \widehat{V}_{1,OLS}} \overset{a}{\sim} \chi^2_1.$$

Require a positive variance difference. A large H supplies evidence of endogeneity; robust comparisons need an appropriate covariance.

Back to why test for endogeneity

Why residual augmentation reproduces the IV coefficient

The fitted and residual parts span the actual regressor

Identity

$y_2 = \widehat{y_2} + \hat{v}_2$, with first-stage residual orthogonal to the instruments.

$$\beta_1 y_2 + \delta_1 \hat{v}_2 = \beta_1 \widehat{y_2} + (\beta_1 + \delta_1)\hat{v}_2.$$

In the augmented regression, $\widehat{y_2}$ is orthogonal to $\hat{v}_2$, conditional on included variables. Its coefficient is the same as the two-stage coefficient.

Why the new error is orthogonal | Back to the residual-augmentation test

Why the new error is orthogonal

The error projection removes the endogenous component

Error decomposition

$u_1 = \delta_1 v_2 + e_1$, with $\text{Cov}(v_2, e_1) = 0$; first stage $y_2 = y_2^* + v_2$.

$$\text{Cov}(z_j, e_1) = \text{Cov}(z_j, u_1) - \delta_1 \text{Cov}(z_j, v_2) = 0.$$

The first term is zero by IV exogeneity; the second by first-stage projection. Since y_2^* is a linear combination of the z_j,

$$\text{Cov}(y_2, e_1) = \text{Cov}(y_2^*, e_1) + \text{Cov}(v_2, e_1) = 0.$$

Back to the component-slope argument

Reading an endogeneity test

Read significance and economic size together

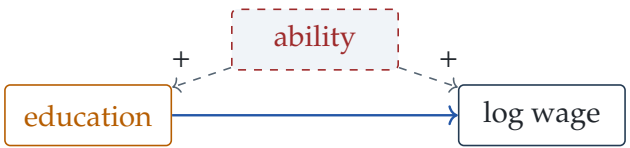
- **Rejection** Evidence of regressor endogeneity, maintaining valid instruments and the model.
- **Nonrejection** The data did not distinguish the estimates sufficiently; assess precision and economic differences.
- **Several suspect regressors** Save each first-stage residual, add them all to the structural regression and test them jointly.

Back to the residual-augmentation test

Why omitted ability can raise the OLS coefficient

Two positive links suggest an upward omitted-variable contribution

$$\log(wage) = \beta_0 + \beta_1 educ + \mathbf{c}'\gamma + \theta abil + e.$$



Ability raises wages; schooling and ability tend to move together after the included variables.

Conditional bias formula | Back to OLS versus IV

The omitted-variable-bias interpretation

The observed estimate difference is consistent with this mechanism

Same full model

$\log(wage) = \beta_0 + \beta_1 educ + \mathbf{c}'\gamma + \theta abil + e.$

$$\operatorname{plim} \hat{\beta}_{OLS} = \beta_1 + \theta \frac{\operatorname{Cov}(\widetilde{educ}, \widetilde{abil})}{\operatorname{Var}(\widetilde{educ})}.$$

The tildes remove the included experience terms. Positive θ and positive covariance of residualized schooling and ability imply an upward contribution.

This mechanism predicts a higher OLS slope when ability raises both schooling and wages.

Back to the OLS and IV comparison

What an overidentification test can establish

Interpret the restrictions jointly for the instrument set

- **Rejection** Review the instrument set: at least one maintained restriction conflicts with the model.
- **Nonrejection** Retain the joint restrictions at the chosen significance level; assess the economic argument for each IV.
- **Exact identification** If $q = 0$, the residual regression has $R^2 = 0$ algebraically. There is no overidentification test.

With heteroskedasticity, use an appropriate robust overidentification test.

Back to the residual-fit test

Three diagnostic questions

Use the test that matches the assumption at issue

Question	Evidence
Are the IVs relevant after controls?	Excluded-instrument first-stage test.
Is the suspect regressor endogenous?	Add its first-stage residual to the structural equation.
Are extra IV restrictions compatible?	Overidentification test using structural residuals.

Back to the residual-fit test

Testing for heteroskedasticity after 2SLS

Local two-parent reproduction; the textbook uses three IVs

Null Intro 2SLS.5: $E(u_1^2 \mid \mathbf{z}) = \sigma^2$.

$$\hat{u}_{i1}^{\,2} = a_0 + a_1 z_{i1} + \cdots + a_m z_{im} + e_i.$$

Test $H_0 : a_1 = \cdots = a_m = 0$ under the additional regularity conditions linked below.

Local two-parent MROZ	Degrees of freedom	p
$F = 3.160$	$(4, 423)$	.0141

The test rejects constant variance. Robust IV SEs require no pretest.

Test conditions | Back to changing error spreads

Conditions behind the auxiliary heteroskedasticity test

A squared-residual test has requirements beyond robust estimation

Auxiliary regression

$\hat{u}_{i1}^{\,2} = a_0 + a_1 z_{i1} + \cdots + a_m z_{im} + e_i.$

Sufficient additional conditions for the auxiliary F test include

$$E(u_1 \mid \mathbf{z}) = 0, \quad E(u_1^{\,4} \mid \mathbf{z}) \text{ constant}, \quad \mathrm{Cov}(\mathbf{x}, u_1 \mid \mathbf{z}) \text{ constant}.$$

Robust IV standard errors do not require a preliminary rejection of constant variance.

Back to changing error spreads

Weighted two-stage least squares

Transform the outcome, regressors and instruments together

Maintain $E(u_i \mid \mathbf{z}_i) = 0$ and $\text{Var}(u_i \mid \mathbf{z}_i) = h_i > 0$. Suppose the variance model can be consistently estimated by $\hat{h}_i$.

$$\frac{y_i}{\sqrt{\hat{h}_i}}, \qquad \frac{x_{ij}}{\sqrt{\hat{h}_i}}, \qquad \frac{z_{ij}}{\sqrt{\hat{h}_i}}.$$

- Divide the intercept too: it becomes $1/\sqrt{\hat{h}_i}$.
- Apply 2SLS to the transformed equation using the transformed instruments.

Back to the changing-spread summary

Theorem 15A.3: efficiency

Efficiency is conditional on the assumptions and comparison class

Theorem 15A.3 Under Assumptions 2SLS.1–2SLS.5, the 2SLS estimator is asymptotically efficient in the class of IV estimators that uses linear combinations of the exogenous variables as instruments.

- This is an asymptotic statement with valid instruments and homoskedasticity.
- Under heteroskedasticity, robust SEs adjust inference; ordinary 2SLS need not be efficient.
- Adding instruments requires additional valid restrictions and may create finite-sample problems.

Back to the changing-spread summary

Noise can halve the OLS slope

Numerical attenuation example: true slope two

$$y = 2x^* + u, \qquad x_1 = x^* + e_1.$$

Assume mutually uncorrelated x^*, u, e_1, with $\mathrm{Var}(x^*) = \mathrm{Var}(e_1) = 1$. What happens to slope two?

$$\operatorname{plim} \hat{\beta}_{OLS} = \frac{\mathrm{Cov}(x_1, y)}{\mathrm{Var}(x_1)} = \frac{2}{1+1} = 1.$$

The observed regressor has twice the true signal's variance, while its covariance with y is unchanged.

Recover the signal with a second report | Back to measurement-error remedies

A separate report recovers the signal

The same example: IV probability limit two

Same model

$y = 2x^* + u, \quad x_1 = x^* + e_1, \quad z_1 = x^* + a_1.$

The components x^*, u, e_1, a_1 are mutually uncorrelated; $\mathrm{Var}(x^*) = \mathrm{Var}(e_1) = 1$. Then $\mathrm{Cov}(z_1, y) = 2$ and $\mathrm{Cov}(z_1, x_1) = 1$.

$$\operatorname{plim} \hat{\beta}_{IV} = \frac{\mathrm{Cov}(z_1, y)}{\mathrm{Cov}(z_1, x_1)} = \frac{2}{1} = 2.$$

The shared true signal links the reports; uncorrelated measurement noise does not enter their covariance.

Back to measurement-error remedies

Two indicators of ability

Separate the latent variable from both test-score errors

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + abil + u. \tag{15.47}$$

$$test_1 = \gamma_1 abil + e_1, \qquad test_2 = \delta_1 abil + e_2, \qquad \gamma_1, \delta_1 > 0.$$

With ability included, schooling is exogenous relative to u. Let $\mathbf{x} = (educ, exper, exper^2)$. The maintained error assumptions include

$$\text{Cov}(e_j, \mathbf{x}) = 0, \quad \text{Cov}(e_j, abil) = 0, \quad j = 1, 2,$$
$$\text{Cov}(e_1, e_2) = 0, \quad \text{Cov}(test_j, u) = 0, \quad \text{Cov}(\mathbf{x}, u) = 0.$$

Substitute the first indicator | Back to measurement-error remedies

Substitute the first ability indicator

Its measurement noise enters the outcome error

Indicator

$test_1 = \gamma_1 abil + e_1$, so $abil = test_1/\gamma_1 - e_1/\gamma_1$.

$$\begin{aligned} \log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 \\ + \alpha_1 test_1 + (u - \alpha_1 e_1), \qquad \alpha_1 = 1/\gamma_1. \end{aligned} \tag{15.48}$$

The new error is $\varepsilon = u - \alpha_1 e_1$. A noisy indicator is included; ability no longer remains entirely omitted.

Which variable needs the IV? | Back to measurement-error remedies

The noisy indicator needs the instrument

Schooling is exogenous under the stated indicator assumptions

Transformed model

$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + \alpha_1 test_1 + \varepsilon$, where $\varepsilon = u - \alpha_1 e_1$.

Under the preceding assumptions,

$$\mathrm{Cov}(educ, \varepsilon) = 0.$$

$$\mathrm{Cov}(test_1, \varepsilon) = -\alpha_1 \mathrm{Var}(e_1), \qquad \mathrm{Cov}(test_2, \varepsilon) = 0.$$

IQ and KWW example | Back to measurement-error remedies

IQ is instrumented by KWW

Example15.6: schooling remains included

Wage equation

$\log(wage) = \beta_0 + \beta_1 educ + \alpha_1 IQ +$ control effects $+ \varepsilon$; IV for IQ: KWW.

KWW is a knowledge-of-the-world-of-work test score. Included controls: experience, tenure, marital status, south, urban and race indicators.

Schooling estimate	Conventional SE	n
.0250	.0166	935

The schooling estimate is small and imprecise. Shared measurement noise could explain the puzzle.

Compare the two ability strategies | Back to measurement-error remedies

Two ability strategies use different assumptions

The outcome error determines which variable needs an instrument

Ability remains omitted	A noisy ability indicator is included
Schooling may correlate with omitted ability in the error. Use an IV for schooling unrelated to that error.	Indicator noise enters the transformed error. Use a second valid indicator as an IV for the first.

An IV that is credible for one equation may fail for the other. Specify the model and error before judging validity.

Back to measurement-error remedies

Example 15.7: is schooling endogenous?

The two-parent MROZ test gives limited evidence

Augmented wage regression

$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + \delta_1 \hat{v}_2 + error.$

The first stage uses both parents and the same experience terms; $n = 428$.

Residual coefficient	SE	t	Approx. p
.0582	.0348	1.67	.095

At 5%, do not reject schooling exogeneity: the sample gives insufficient evidence of endogeneity. Report the OLS and IV estimates.

Back to endogeneity-test sequence

When would error spreads vary?

Different observations can have different conditional uncertainty

$$\mathrm{Var}(u_1 \mid \mathbf{z}) = h(\mathbf{z}).$$

- **Conditioning variables** In the wage example, $\mathbf{z}$ includes experience and the parental-education instruments.
- **Wage uncertainty** Workers with different experience can have different error spreads, even after allowing experience to affect the fitted mean.
- The mean relation can remain correctly specified while the error spread changes with these exogenous variables.

Optional heteroskedasticity test

Back to changing-spread summary

MROZ: the coefficient stays fixed

The covariance choice changes uncertainty for the same model

Schooling coefficient	Conventional	HC1 robust
Estimate	.06140	.06140
SE	.03144	.03334

Same 428 observations, included variables and instruments. Robust covariance changes the standard error, not the coefficient.

In Stata, add `vce(robust) small` to the earlier IV command; `small` applies the degrees-of-freedom correction.

Read the robust interval | Back to changing-spread summary

MROZ: read the robust interval

An interval describes uncertainty under the maintained IV assumptions

Two-parent model

$\hat{\beta}_1 = .06140, \quad \text{se}_{HC1} = .03334.$

$$.06140 \pm 1.96(.03334).$$

Approximate 95% interval: $[-.0039, .1267]$.

The interval includes zero. Robust SEs may be larger or smaller than conventional SEs.

Why error spread can vary | Back to changing-spread summary

A weighted alternative

Downweight observations with larger conditional error variance

Two distinct tasks
Robust standard errors allow changing error spreads when calculating uncertainty.

Weighted 2SLS additionally changes the estimation weights: observations with larger estimated error variance receive less weight.

The variance model needs justification. Transform the outcome, explanatory variables and instruments together.

Optional weighted-2SLS procedure | Optional efficiency theorem

Back to changing-spread summary

Attenuation with included variables

Use the signal remaining after included variables

True model

$y = \beta_0 + \beta_1 x_1^* + \beta_2 x_2 + u$; observed $x_1 = x_1^* + e_1$.

Tildes remove the linear prediction on $1, x_2$. Maintain classical noise and true-regressor exogeneity.

$$\operatorname{plim} \hat{\beta}_{1,OLS} = \beta_1 \frac{\operatorname{Var}(\widetilde{x_1^*})}{\operatorname{Var}(\widetilde{x_1^*}) + \operatorname{Var}(e_1)}.$$

Other coefficients need not attenuate toward zero.

Back to the attenuation formula

Further IV applications

Optional research examples beyond the main lecture

El Niño and trade: short overview

Pollution and exports

Appendix contents | Lecture takeaways

Climate shocks can synchronize harvest risks

El Nino and trade — shared shocks can limit gains from trade

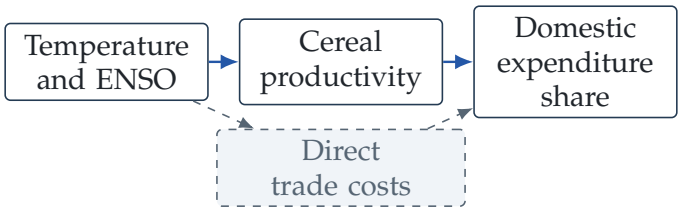
- **Economic setting**: countries can use trade to share local cereal-production risks.
- **Complication**: when climate shocks move harvests together, trading partners may be hit at the same time.
- Dingel, Meng and Hsiang study how cereal productivity and its global spatial correlation relate to domestic expenditure shares.

Temperature and ENSO supply candidate productivity variation; a physical shock still needs an outcome-specific exclusion argument.

Back to lecture takeaways | IV argument

Climate variation still needs an exclusion argument

El Nino and trade — relevance and validity remain separate



Know the restriction: climate must not shift domestic shares through unmodeled trade-cost pathways.

Back to lecture takeaways | El Nino summary

Pollution can reduce export revenue

He, Sheng and Zhang — a recent study of Chinese manufacturers

$$\log(exports) = \beta_0 + \beta_1 \log(PM_{2.5}) + \text{included factors} + u$$

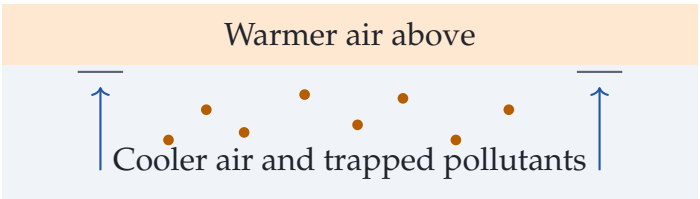
- Chinese firm-product export observations, 2000–2007.
- More production can generate more pollution; local economic conditions can also move both variables.
- The study uses thermal inversion days as an instrument for pollution.

He, Sheng and Zhang, Journal of Public Economics (2026).

Back to lecture takeaways | Inversion mechanism

A warm upper layer traps ground-level pollution

The inversion mechanism supplies first-stage relevance



Mechanism: warmer air limits vertical dispersion.

Back to lecture takeaways | First-stage evidence

Inversion days predict pollution

The first stage addresses relevance

$$\log(PM_{2.5}) = \pi_0 + \pi_1 \log(\textit{inversion days}) + \text{included factors} + v$$

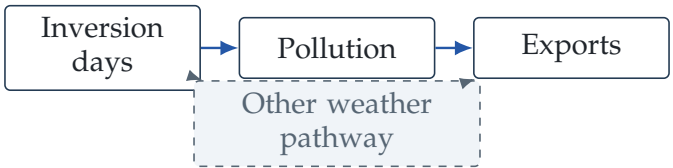
- More inversion days predict higher pollution after the included adjustments.
- The published first-stage coefficient is 0.0653, with standard error 0.007.

Back to lecture takeaways | Exclusion

A physical instrument still requires exogeneity

Weather and other pollutants are concrete exclusion concerns

$$\log(exports) = \beta_0 + \beta_1 \log(PM_{2.5}) + \text{included factors} + u$$



Weather and copollutant pathways need scrutiny.

Back to lecture takeaways | Published estimate

Interpret the IV estimate in percentage units

Published Table 2, column 2 — separate relevance from validity

IV export-value specification	Estimate (SE)
First stage: log(*inversion days*)	0.0653 (0.007)
Export equation: log($PM_{2.5}$)	−0.9179 (0.410)
Observations	3,046,936

- Under the assumptions, 1% higher pollution implies about 0.92% lower export revenue.
- Relevance uses the first stage; exclusion uses the institutional argument.

Back to lecture takeaways | Pollution and exports